

EE 471C / EE 38K-17 Wireless Communications Lab – Fall 2026

Homework #1

This homework focuses on researching current wireless technologies and trends, including emerging 6G developments. It is less mathematical than future assignments and should be completed independently. Typed responses are recommended for clarity.

Wireless Devices Comparison

Fill in the following table for three current flagship cellular devices:

	Samsung Galaxy S25 Ultra	iPhone 17 Pro Max	Google Pixel 10 Pro
What wireless technologies are supported (e.g., WiFi 7, 5G, Bluetooth)?			
Which cellular standards are supported (e.g., 5G NR, LTE, etc.)?			
Which frequency bands are used for communication (e.g., mmWave, sub-6 GHz)?			
What is the maximum (claimed) data rate supported via cellular?			
What is the maximum (claimed) data rate supported via WiFi?			
What operating system is employed?			

Cellular Provider Comparison

Choose two major cellular providers in Austin and fill in the following table:

	Provider A	Provider B
What cellular standards are supported (e.g., 5G SA, 5G NSA, LTE)?		
What is the monthly charge for an unlimited voice plan?		

What is the monthly charge for a data plan with at least 10 GB?		
What is the cost of voice calls when roaming in Germany (no international plan)?		
What is the cost of data when roaming in Germany (no international plan)?		

6G Technology Exploration

Answer the following questions based on your research into 6G wireless technologies:

- What are the expected capabilities of 6G compared to 5G (e.g., latency, data rate, coverage)?
- What frequency bands are being considered for 6G (e.g., THz spectrum)?
- What are some potential applications of 6G (e.g., holographic communication, digital twins, tactile internet)?
- What challenges must be overcome to implement 6G (e.g., hardware, energy efficiency, spectrum availability)?
- Which organizations or countries are leading 6G research and development?
- Are there any prototype devices or testbeds currently being used for 6G trials?

Exploration Question

On a separate page, describe one question about wireless communication that you would like to see answered in class. This could relate to emerging technologies (e.g., 6G, satellite-based mobile networks), security and privacy, IoT, edge computing, or AI/ML applications in wireless communication.